

An Instructor Ratings Analysis: How Relationships Among Ratings and Factors Associated with Teaching Quality Based

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1 Introduction

Universities commonly assess teaching quality through student surveys. However, these evaluations may reflect a range of factors beyond an instructor's teaching ability. This study utilizes student evaluation data from a large American university to explore the relationships among the five rating dimensions.

2 Relationships Among the Ratings

Overall, STEM instructors received the lowest mean ratings across all five dimensions, particularly in easiness (2.87), suggesting that STEM students have relatively higher difficulty and are less satisfied with the instructor quality. In contrast, Pre-prof instructors received the highest average ratings, helpfulness, and easiness, but the lower rate for average interest of only 3.23. Thus, instructor quality ratings may not necessarily correspond to greater student interest.

Notably, the standard deviations were similar across the four disciplines, which suggest students tend to come out with similar conclusion. But for easiness, it showed more noticeable variation, which may indicate that the level of difficulty of the subject tend to be vary.

As shown in Figure 1, all five rating dimensions were positively correlated. Quality was strongly associated with helpfulness and clarity. In contrast, quality showed weaker correlations with easiness and raterInterest. The detailed analysis of the correlation will be further explored in the subsequent analysis.

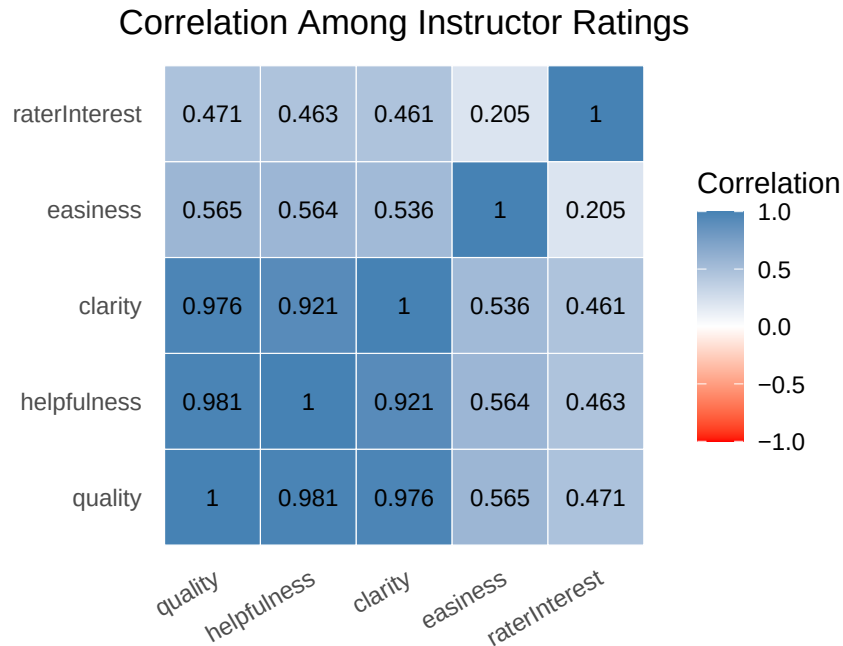


Figure 1

3 Model

We first examined the distributions of numerical variables using histograms and found that quality was slightly left-skewed, while easiness and raterInterest were relatively symmetric. We then used scatterplots to explore their relationships with quality. Both easiness and raterInterest showed positive and approximately linear relationships, but no clear patten between numYears and quality. Given that quality is a continuous variable, we adopted a Gaussian linear regression, with its assumptions to be assessed through residual diagnostics.

After performing diagnostic plots, the results suggest that the model performs reasonably well, with no major concerns about linearity or normality. Although the Scale-Location plot shows slightly spreaded, but the Breusch-Pagan test found no significant evidence of heterogeneity with $p = 0.133$. We also identified 14 potentially influential observations by Cook's distance, while sensitivity analysis showed stable results after be excluded. Meanwhile, the residuals showed no clear pattern across observations, with VIF 1. So, multicollinearity and independent are satisfied. Based on these findings, our final model is below.

$$\begin{aligned}
 \text{Quality}_i \mid \mu_i, \sigma &\sim \text{Normal}(\mu_i, \sigma) \\
 \mu_i &= \beta_0 + \beta_1 \cdot \text{gender}_i + \beta_2 \cdot \text{numYears}_i \\
 &\quad + \beta_3 \cdot \text{PreProf}_i + \beta_4 \cdot \text{SocSci}_i + \beta_5 \cdot \text{STEM}_i \\
 &\quad + \beta_6 \cdot \text{easiness}_i + \beta_7 \cdot \text{raterInterest}_i + \beta_8 \cdot \text{pepper}_i
 \end{aligned}$$

4 Statistical Significance and Practical Importance

As shown in Table 1, easiness, student interest, pepper, and STEM discipline were positively associated with instructor quality after holding for other predictors ($p < 0.05$). The association with pepper was the most noticeable, as instructors treated as attractive received approximately 0.56 points higher in ratings than those who were not, with other variable fixed. And student interest and easiness also showed substantial associations, with each one-point increase corresponding to 0.54 and 0.51 higher in quality rating, respectively. In contrast, gender and numYears do not show statistically significant associations ($p > 0.05$). The confidence intervals for the significant predictors excluded zero, further supporting these finding of associations.

Table 1: Multiple Linear Regression Results for Instructor Quality

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.181	0.242	[-0.657, 0.296]	0.457
gender	0.047	0.065	[-0.081, 0.174]	0.472
numYears	0.018	0.010	[-0.002, 0.037]	0.081

Table 1: Multiple Linear Regression Results for Instructor Quality

Predictor	Estimate	SE	95% CI	p-value
disciplinePre-prof	0.097	0.091	[-0.083, 0.276]	0.291
disciplineSocSci	0.019	0.089	[-0.156, 0.193]	0.834
disciplineSTEM	0.295	0.081	[0.135, 0.455]	<0.001
easiness	0.513	0.042	[0.429, 0.596]	<0.001
raterInterest	0.544	0.059	[0.427, 0.661]	<0.001
pepper	0.562	0.099	[0.366, 0.757]	<0.001

5 Quality and Discipline

Although STEM instructors received the lowest average quality ratings, but after holding other predictors constant the situation changed. The STEM instructors were estimated to receive approximately 0.30 points higher than Humanities instructors. In comparison, neither Pre-Prof nor Social Science showed statistically significant differences from Humanities after adjustment. This reversed result may reflect that the lower quality ratings are more closely related to course easiness and personal interest in STEM courses but not just discipline it self.

A Appendix 1

```
library(dplyr)
library(knitr)

# Select the five rating variables
ratings <- c("quality", "helpfulness", "clarity", "easiness", "raterInterest")

# Summary statistics by discipline
discipline_summary <- data %>%
  group_by(discipline) %>%
  summarise(
    N = n(),
    across(all_of(ratings), ~sprintf("%.2f (%.2f)", mean(.x, na.rm = TRUE),
      sd(.x, na.rm = TRUE)))
  )

# Display the table
kable(discipline_summary,
      col.names = c("Discipline", "N", "Quality", "Helpfulness", "Clarity",
        "Easiness", "Rater Interest"),
      caption = "Instructor Ratings by Discipline (Mean and SD)",
      align = "lcccccc")
```

```
#histogram
variables <- c("quality", "numYears", "easiness", "raterInterest")
dist_data <- data %>%
  pivot_longer(all_of(variables), names_to = "Variable", values_to = "Value")

# let review distributions
ggplot(dist_data, aes(x = Value)) +
  geom_histogram(bins = 20, fill = "black", color = "white") +
  facet_wrap(~Variable, scales = "free") +
  labs(title = "Distribution of Numerical Variables", x = "Value",
    y = "Frequency") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))
```

```
scatter_data <- data[, c("quality", "numYears", "easiness", "raterInterest")]
```

```

scatter_long <- pivot_longer(scatter_data, -quality,
                             names_to = "Variable", values_to = "Value")

# Plot relationships with quality
ggplot(scatter_long, aes(x = Value, y = quality)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  facet_wrap(~Variable, scales = "free_x") +
  labs(title = "Relationships Between Numerical Predictors and Quality",
       x = "Predictor Value", y = "Quality") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))

```

```

ratings <- data[, c("quality", "helpfulness", "clarity", "easiness", "raterInterest")]

cor_matrix <- cor(ratings)
cor_long <- melt(cor_matrix)

# Plot correlation heatmap
ggplot(cor_long, aes(Var1, Var2, fill = value)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(value, 3)), size = 3) +
  scale_fill_gradient2(low = "red", mid = "white", high = "steelblue",
                      midpoint = 0, limits = c(-1, 1)) +
  labs(title = "Correlation Among Instructor Ratings",
       x = NULL, y = NULL, fill = "Correlation") +
  coord_fixed() +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.text.x = element_text(angle = 30, hjust = 1),
    panel.grid = element_blank()
  )

```

```

model1 <- lm(quality ~ gender + numYears + discipline + easiness + raterInterest + pepper, data = ratings)
summary(model1)

# Test regression assumptions
par(mfrow = c(2, 2))
plot(model1)
par(mfrow = c(1, 1))

```

```

plot(residuals(model1),
     xlab = "Observation Order",
     ylab = "Residuals",
     main = "Residuals vs Observation Order")
abline(h = 0, col = "red")

# Test multicollinearity
vif(model1)

library(lmtest)

# Test heteroscedasticity
bptest(model1)

# Test influential observations
sort(cooks.distance(model1), decreasing = TRUE)[1:10]
cooks <- cooks.distance(model1)
which(cooks > 4/nobs(model1))

# Identify potentially influential observations
influential <- cooks > 4 / nobs(model1)

# check and refit model without flagged observations
model_sensitivity <- lm(
  formula(model1),
  data = model.frame(model1)[!influential, ]
)

round(cbind(
  Original = coef(model1),
  Sensitivity = coef(model_sensitivity)
), 3)

# Summarize regression results
model_table <- tidy(model1, conf.int = TRUE) %>%
  mutate(
    across(c(estimate, std.error, conf.low, conf.high), ~round(.x, 3)),
    CI = sprintf("[% .3f, % .3f]", conf.low, conf.high),
    p.value = ifelse(p.value < 0.001, "<0.001", sprintf("%.3f", p.value))
  ) %>%
  select(term, estimate, std.error, CI, p.value)

```

```
# Display regression table
kable(model_table,
      col.names = c("Predictor", "Estimate", "SE", "95% CI", "p-value"),
      caption = "Multiple Linear Regression Results for Instructor Quality",
      align = "lcccc")
```

B Appendix 2: AI Use Disclosure

This report used ChatGPT to assist with R coding, data visualization and part of regression diagnostics. Representative prompts included “How should I assess regression assumptions?”, “How should I check the influential observation and to assess whether we should remove them” and “Help me to create a heatmap for correlation.” AI-generated suggestions were reviewed, modified, and checked against the actual R outputs and the each line of code true meaning before being incorporated into the analysis and report.